Mrityunjoy Saha

Director's Fellow

Indian Institute of Technology Jodhpur, Rajasthan, India

CFD | Mathematical Modeling | Numerical Simulations

Personal Email · Institute Email · Phone

LinkedIn · Google Scholar · ORCID

SUMMARY

- Computational fluid dynamics researcher with experience in numerical simulation of complex flow systems using spectral collocation methods and scientific computing. Having experience working in collaborative environments. Skilled in C/C++, Python, MATLAB, and HPC-based computations. With publications in leading journals such as *Physics of Fluids* and *Computers & Fluids*. Interested in applying mathematical modeling and simulation techniques to engineering problems in microfluidics, aerospace systems, and autonomous technologies.

EDUCATION

- **Ph.D. in Fluid Mechanics** **2021–2025**
  Indian Institute of Technology Jodhpur, India
  Thesis: Hydrodynamic Stability Analysis on a Class of Complex Flow Systems (Submitted)
  CGPA: 8.69/10

- **M.Sc. in Mathematics (Applied Mathematics)** **2018-2020**
  Cooch Behar Panchanan Barma University, West Bengal, India
  CGPA: 8.19/10

- **B.Sc. (Hons.) in Mathematics** **2015-2018**
  Dinhata College, Cooch Behar Panchanan Barma University, West Bengal, India
  Percentage: 62.5

SKILLS

- **Programming Languages**
  C/C++, Python

- **Scientific Computing & Simulation**
  MATLAB, Mathematica, COMSOL Multiphysics, High Performance Computing (HPC)

- **Domain Expertise**
  Computational Fluid Dynamics (CFD), Hydrodynamic Stability, Porous Media Flows, Numerical PDEs.

PUBLICATIONS

- **M. Saha** and S. Ghosh, "On the magnetohydrodynamic instability pattern of an anisotropic porous media flow in a rotating channel," Physics of Fluids, vol. 38(1), pp. 014122, 2026.

- M. M. Hossain, **M. Saha**, S. Ghosh, and H. Behera, "Stability dynamics of an odd viscosity induced fluid flow over a vibrating inclined surface," Physics of Fluids, vol. 37(10), pp. 104108, 2025.

- **M. Saha**, S. Sengupta, S. Mukhopadhyay, and S. Ghosh, "Flow instability in a rotating channel loaded with an anisotropic porous material," Computers & Fluids, vol. 299, pp. 106689, 2025.

ONGOING RESEARCH

- **M. Saha**, S. Ghosh, M. Roy, and M. Mishra, "Modal and non-modal stability of Poiseuille flow through anisotropic inhomogeneous porous materials." (Under review)
- **M. Saha**, S. Ghosh, and K. C. Sahoo, "Dual instability in a vertical crossflow-induced oscillating channel flow." (In preparation)
- S. Panda, **M. Saha**, and S. Ghosh, "Effect of uniform vertical crossflow on rotational instability in a channel flow." (In preparation)

EXPERIENCE

- **Teaching Assistant** **2022–Present**
  Indian Institute of Technology Jodhpur
  Courses assisted: Mathematics I & II, Scientific Computation, Mathematical Modeling, Data Science, Applied Mathematics & Statistics, Linear Algebra, Complex Analysis.

PROFESSIONAL TRAINING

- **Python Programming for Computational Fluid Dynamics Applications** **June-July, 2024**
  Electronics & ICT Academy, Indian Institute of Technology Guwahati

CONFERENCE PRESENTATIONS

- **78th Annual Meeting of the APS Division of Fluid Dynamics (APS DFD 2025)**,
  George R. Brown Convention Center, Houston, Texas, USA.
- **International Conference on Recent Advances in Fluid Mechanics and Nanoelectronics 2023**,
  Manipal Institute of Technology, Bengaluru, India.

AWARDS AND SCHOLARSHIPS

- Director's Fellowship, Indian Institute of Technology Jodhpur.
- American Physical Society - Division of Fluid Dynamics Travel Award 2025.
- ANRF International Travel Support for APS DFD 2025.
- CSIR Foreign Travel Grant for APS DFD 2025.
- Qualified Graduate Aptitude Test in Engineering (GATE), Mathematics (2022, 2023).
- DST INSPIRE Scholarship - Government of India.
- SVMCM Scholarship (UG, PG, and Higher Secondary).
- Sitaram Jindal Scholarship for B.Sc.

REFERENCE

- **Dr. Sukhendu Ghosh** (Ph.D. Supervisor)
  Associate Professor, Department of Mathematics, IIT Jodhpur
  Email: sukhendu.math@iitj.ac.in